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I N D E X

Sr No.	Experiment Description	Experiment Date	Submission Date	Remarks / Signature
01.	Co 1: Vehicle Dynamics.			
02.	Co 2: Elements of Automobile.			
03.	Co 3: EV power Transmission System			
04.	Co 4: Vehicle Control Unit (VCU)			
05.	Co 5: EV Charging Technologies.			

COs 1:Vehicle Dynamics

Introduction:

The vehicle movement is concerned with the different type of forces which act on a vehicle when it propels. Newton's second law of motion played the important role in vehicle movement. According to Newton's second law "the acceleration of an object is proportional to the net force ~~ext~~ exerted on it".

Hence a vehicle accelerates when the net force acting on it is not zero. In a vehicle several forces act on it and the net or resultant force governs the motion according to the Newton's second law.

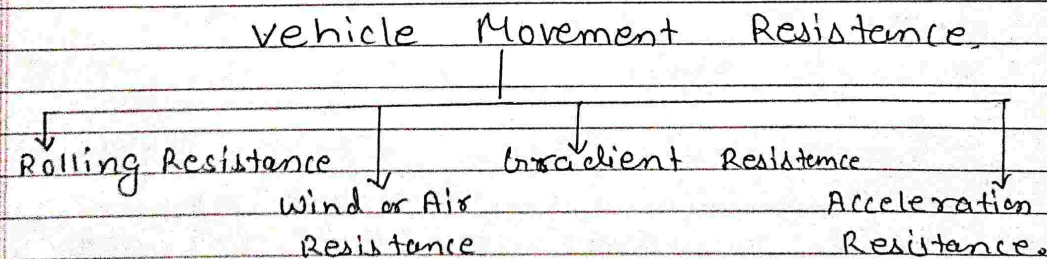
The propulsion unit of the vehicle delivers the force necessary to move the vehicle forward. The force of the propulsion unit helps the vehicle to overcome the resisting forces due to gravity, air and tire resistance.

The acceleration of the vehicle depends on :-

- a. the power delivered by the propulsion unit.
- b. the road condition.
- c. the aerodynamics of the vehicle.
- d. the composite mass of the vehicle.

Vehicle Movement Resistance :-

- A moving vehicle opposed by various external forces known as resistances. A moving vehicle has to overcome the all resistance. There are following types of resistance for moving vehicle.



1. Rolling Resistance (R_g):

- Rolling resistance is friction between tyre and road surface while tyre rolling.

→ Rolling resistances depend on the following parameters.

- ① Road Characteristics.
- ② Tyre Characteristics.
- ③ Weight of the vehicle.
- ④ Vehicle Speed.

→ The value of rolling resistance is determined by:

$$(R_r) = K_R \cdot W = K_R \cdot m \cdot g$$

Where,

K_R = Constant of rolling resistance.

W = weight of vehicle.

m = mass of vehicle in kg.

g = Acceleration due to gravity = 9.8 m/s^2

2. Wind or Air resistance.

- It is the resistance offered by air to the vehicle motion. It depends on the following factors:

- ① Size of vehicle.
- ② Shape of vehicle.
- ③ Speed of vehicle.
- ④ Wind velocity.

Air resistance,

$$R_A = \left(\frac{1}{2} \rho C_d\right) A \cdot v^2$$

ρ = Air density in kg/m^3 .

C_d = Coefficient of drag.

A = Cross sectional area.

V = Vehicle speed.

$$K_A = \text{Coefficient of Air resistance} = \frac{1}{2} \rho C_d$$

$$\therefore \text{Air resistance } R_A = K_A \cdot A \cdot v^2 \text{ Km/h}$$

3. Gradient Resistance:

The gradient resistance is due to slope or steepness of road resistance. If the vehicle is loading uphill at a slope of θ , then the weight of the vehicle, w has two components: One perpendicular to the road surface (with a value $w \cos \theta$) and the other along the road surface (with a value $w \sin \theta$).

The gradient resistance is given by $R_g = w \sin \theta = mg \sin \theta$

w = weight of vehicle.

m = mass of vehicle in kg,

g = Acceleration due to gravity 9.8 m/s^2 .

COs:2

Elements of Automobile.

Introduction:

An automobile is a self-propelled motor vehicle intended for passenger transportation on land. It usually has four wheels and internal combustion engine fueled most often by gasoline, a liquid petroleum product.

The element of automobile are as following.

1. Suspension and Damping System.
2. Brake System.
3. Chassis
4. Tyre
5. Lighting System.
6. AC or DC motor.
7. Air and Heat control system.
8. Vehicle Control Unit.
9. Gear Box
10. Battery Charger.
11. Power Control Unit.

1. Suspension and Damping System.

Suspension: System of tires, tires air, springs, shock absorbers and linkages that connects a vehicle to its wheels and allows relative motion b/w two.

Need of suspension:

- (i) For absorbing shocks and vibration caused due to road irregularities.
- (ii) For transmitting vehicle load to wheels.
- (iii) For monitoring the stability of vehicle (contact of wheels to ground).
- (iv) For providing cushioning and ride comfort to passengers.
- (v) For preventing body squat and body dive.

2. Brake System:

A brake is a mechanical device whose function is used to absorb the

kinetic energy of system by means of friction.

3. Chassis:

Chassis is a term that should be used to refer to the load-bearing part of a car's frame.

4. Tyre

Tyre (or tire) is a circular and ring like part of a vehicle which comes in contact with ground. Tyres are fitted on rims and are filled with compressed air.

5. Lighting System.

Automotive exterior lighting serves to ensure safety in road traffic.

6. AC or DC motor.

There are a wide variety of Propulsion System available or potentially available for automobiles and other vehicles. An automotive electric motor is an electrical machine that converts electrical energy into mechanical energy.

7. VCU (vehicle Control Unit).

VCU is abbreviations for vehicle control unit. As a powertrain domain controller, the vehicle control unit (VCU) can provide torque coordination, operation and gearshift strategies.

8. Gear box.

All electric car are automatic. In addition plug-in hybrids also use automatic transmission. In short, the design of these vehicle means they don't require gears, as the single-gear transmission is simple and efficient.

9. Battery Charger.

A device specifically designed to supply electrical power for recharging plug-in electric vehicles.

10. Power Control Unit.

Electric power control unit integrates a number of devices such as the inverter, the low voltage converter, and the vehicle control unit (VCU) to manage the flow of power within the electric vehicle.

COs 3:

EV Power Transmission System.

Transmission System.

Transmission system for electric vehicles includes:

- (i) A gear box which provides different rotation between a drive motor and drive wheels during running.
- (ii) A differential which serves to absorb a power of gearbox to overcome rotation speed diff. b/w the drive wheel.
- (iii) Two axles which are used to deliver the power to drive wheels from differential.
- (iv) A drive shaft which delivers power from gearbox.

In, EV, transmission is of two types:

- (i) Single-speed transmission.
- (ii) Multi-speed transmission.

EV Power Train

Powertrain refers to set of components that generate power required to move the vehicle and deliver it to wheels.

The power train of EV is simpler system, comprising of far fewer components than the vehicle powered by an internal combustion engine.

An EV powertrain has 60% fewer components than that of an ICE vehicle. The components are described as:-

- (i) Battery Pack : It is made up of multiple lithium-ion cells and stores the energy needed to run vehicle.
- (ii) DC-AC converter: The DC supplied by battery pack is converted to AC and supplied to electric motor.
- (iii) Electric Motor - Converts electrical energy to mechanical energy.
- (iv) On-Board Charger - Converts AC received to charge port to DC & control the amount of current flowing into the battery.

pack.

Battery Parameters:

Consider, a battery of 48v with a Capacity (C) of 15kwh. Various parameters of battery are as following.

A. Voltage

The term "battery voltage" refers to the difference in electrical potential between the positive and negative terminals of a battery.

B. Current

When a load is connected between both the positive and negative terminal of battery then electric current flow through the load.

C. Battery Efficiency.

Battery efficiency is defined as the amount of energy we can get out of a battery relative to the amount of energy that's put into it.

The ideal value of battery efficiency is 80-90%.

D. SOC (State of Charge):

The state of charge is describe the amount of energy available in a battery.

E. State of health (SOH).

The battery state of health is a measurement that includes indicates the level of degradation and remaining capacity of the battery.

F. Depth of Charge.

Depth of Charge (DoD) is Figure of merit that is often used instead of SOC.

G. Operating Temperature.

The operating temperature of a battery is the value of that temperature at which battery safely charging and discharging.

E. Life Cycle.

The life cycle of Li-ion battery is affected by the depth of charge.

COs 4:

Vehicle Control Unit (VCU)

Introduction:

VCU - is an electronic control unit that manages a vehicle subsystem. It is a major controlling part of whole electric vehicle.

The different ECUs ideally involve components like motor controller, BMS, On Board charger, etc.

VCU usually interact, aggregate data and then make decisions. A VCU communicates with these ECUs using various communication protocols such as CAN, LIN and Ethernet.

VCU ensures that vehicle operates efficiently and safely by monitoring critical systems such as battery health, motor temp. and state of charge.

Working OF VCU:

The VCU in automotive is divided into three categories that are as following.

1. Powertrain Control Unit.

It regulates the ignition, fuel supply, valve timing, emissions, turbo boost pressure, idle speed, throttle position and cruising speeds.

2. Body Control Unit

When communicating with other ECUs in car, the BCM relies on vehicles bus system (CAN, LIN, etc) and serves as a hub or gateway.

3. Chassis Control Module.

Maintain the integrity of vehicles key safety systems. It collects information from different suspension, steering and braking sensors.

Having, a VCU for each systems operation helps simplify or organize the whole things.

BMS:

To maintain safety operations of batteries they require a protective device to be built into each pack is called BMS

It makes decisions on charge and discharge rates on the basis of load demands, cell voltage, current and temp. measurements, estimated battery, SOC, Capacity, impedance and many more things.

Need of BMS

1. Detects unsafe operating condition and responds.
2. Protects cells of battery from damage in abuse and failure cases.
3. Prolongs life of battery.
4. Maintains battery in a state.

Functions of BMS:

1. Sensing and high voltage control:
Measure voltage, current, temp., thermal management.
2. Protection - Over - charge, over-discharge, Over current, short circuit, etc.
3. Interface - Range, estimation, communications, data recording, reporting.
4. Performance Management: State of charge estimation power limit computation, balance and equalize cells.
5. Diagnostics - Abuse detection, SOH estimation, state of life (SOL) estimation.

C.O.s 5:

EV Charging Technologies

Introduction to V2G:

The basic concept of V2G power is that EVs provide power to the grid while they are parked.

EDV (electric drive vehicle) can be a battery - electric vehicle, hybrid or fuel cell vehicle connected to grid.

Each vehicle must have three required elements for V2G:

1. A power connection to the grid for electrical energy flow.
2. Control or logical connection necessary for comm. with grid operators.
3. Precision metering on-board the vehicle.

Benefits :

1. Consumer may profit from the use of EVs because electricity is cheaper than petrol.
2. EVs in a V2G configuration could provide additional revenue to owners who wish to sell power back to grid.
3. It helps in achieving sustainability targets.
4. The energy stored in powerpacks can be used in power outage, continuous operation is guaranteed.

Applications :

Applications can be grouped into "Customer applications" that focuses on services to electricity customer or EV owner/operator and "System applications" that focus on ancillary services to grid.

EV battery can be used to temporary supply homes (V2H) and buildings (V2B) back-up power during outages.

Bidirectional EV Charging System.

EVs with bidirectional (two-way) charging capability can be used to power a home, feedback energy into electricity grid and even provide back up in the event of emergency.

An EV is essentially a large battery on wheels, so bidirectional chargers can enable and reduce electricity costs. This technology is V2G.

A bi-directional charger is an advanced EV charger capable of 2-way charging.

Bi-directional chargers operate much like an inverter and convert AC to DC during charging and the reverse during discharge.

It can work only with vehicles that are compatible with 2-way DC charging.

Uses of Bidirectional Charging:-

1. V2G → To send and export energy into the electricity grid when the demand is high. (JP thousands of vehicles with V2G tech. are plugged in and enabled this has potential to transform how electricity is stored and generated on massive scale.

2. V2H → EV to be used much like home that battery system to store energy & power our home.

Ex - Tesla Powerwall
(Capacity 13.5 kWh)

Ex. of EVs with Bidirectional Charging - Nissan Leaf ZEI, Hyundai, Kia EV6, etc.

→ Some bidirectional chargers are -

Wall box, Quaser, Highbury, Fermato FE-15 are Universal chargers for bi-directional DC charges.